OPTIMISATION

1. a) The Lagrangian of the problem is

$$L(x,\lambda) = x_1 - x_2 + 2x_3 + \lambda_1(x_1 + x_2 + x_3) + \lambda_2(x_1^2 + x_2^2 - 1).$$

The necessary conditions of optimality are

$$0 = \frac{\partial L}{\partial x_1} = 2x_1\lambda_2 + \lambda_1 + 1, \qquad 0 = \frac{\partial L}{\partial x_2} = 2x_2\lambda_2 + \lambda_1 - 1, \qquad 0 = \frac{\partial L}{\partial x_3} = \lambda_1 + 2,$$

$$0 = x_1 + x_2 + x_3, \qquad 0 = x_1^2 + x_2^2 - 1.$$

Solving these equations yields the candidate optimal solutions

$$x_1 = -\frac{\sqrt{10}}{10}, \quad x_2 = -3\,\frac{\sqrt{10}}{10}, \quad x_3 = 4\,\frac{\sqrt{10}}{10}, \quad \lambda_1 = -2, \quad \lambda_2 = -\frac{\sqrt{10}}{2},$$

$$x_1 = \frac{\sqrt{10}}{10}, \quad x_2 = 3\,\frac{\sqrt{10}}{10}, \quad x_3 = -4\,\frac{\sqrt{10}}{10}, \quad \lambda_1 = -2, \quad \lambda_2 = \frac{\sqrt{10}}{2}.$$

[6 marks]

b) Note that the Lagrangian can be rewritten as

$$L(x,\lambda) = (x_1 + \lambda_1 x_1 + \lambda_2 x_1^2) + (-x_2 + \lambda_1 x_2 + \lambda_2 x_2^2) + (2x_3 + \lambda_1 x_3) - \lambda_2.$$

The third term is bounded from below only if $\lambda_1 = -2$. Note then that the first two terms can be minimised only if $\lambda_2 > 0$, yielding the minimizers

$$x_1^\star = \frac{1}{2\lambda_2}, \qquad x_2^\star = \frac{3}{2\lambda_2},$$

in which we have used $\lambda_1 = -2$, hence

$$x^\star(\lambda) = x^\star(\lambda_2) = \begin{bmatrix} \frac{1}{2\lambda_2} & \frac{3}{2\lambda_2} & x_3 \end{bmatrix}$$

where x_3 is arbitrary. [8 marks]

c) Replacing $x^\star(\lambda)$ into the constraints yields

$$0 = \frac{2}{\lambda_2} + x_3, \qquad 0 = \frac{5}{2\lambda_2^2} - 1,$$

hence $\lambda_2 = \pm\frac{\sqrt{10}}{2}$ and $x_3 = \mp 4\frac{\sqrt{10}}{10}$. Note now, from part b), that λ_2 has to be positive, otherwise one would not be able to minimise the Lagrangian. As a result, the only pair $(x^\star, \lambda^\star)$ which satisfies the Lagrangian Sufficiency Theorem is

$$x^\star = \left(\frac{\sqrt{10}}{10},\ 3\,\frac{\sqrt{10}}{10},\ -4\,\frac{\sqrt{10}}{10}\right) \qquad \lambda^\star = \left(-2,\ \frac{\sqrt{10}}{2}\right).$$

This coincides with one of the candidate solutions determined in part a).

[6 marks]

Answers

2. a) Solving the constraint for x_2 yields $x_2 = r - x_1 = 2 - x_1$, which replaced into the cost yields

$$f_{red}(x_1) = f(x_1, 2-x_1) = \frac{x_1}{2-x_1} + \frac{2-x_1}{x_1-1}.$$

Differentiating with respect to x_1 yields the necessary condition

$$0 = \frac{df_{red}}{dx_1} = \frac{x_1^2 - 2}{(2-x_1)^2(x_1-1)^2}$$

from which we obtain the only (recall that $x_1 \geq 0$) candidate optimal solution $x_1 = \sqrt{2}$, giving $x_2 = 2 - \sqrt{2}$ (which is also feasible, since $0 \leq x_2 < 1$). To verify local optimality note that

$$\frac{d^2 f_{red}}{dx_1^2} = 2\frac{x_1^3 - 6x_1 + 6}{(2-x_1)^3(x_1-1)^3}$$

which evaluated at $x_1 = \sqrt{2}$ gives $\frac{d^2 f_{red}}{dx_1^2}(\sqrt{2}) \approx 48 > 0$, hence the point $(\sqrt{2}, 2 - \sqrt{2})$ is the optimal solution. [4 marks]

b) The Lagrangian of the reduced problem is

$$L(x_1, x_2, \lambda) = \frac{x_1}{2-x_1} + \frac{x_2}{1-x_2} + \lambda(2 - x_1 - x_2)$$

and the necessary conditions of optimality are

$$0 = \frac{dL}{dx_1} = -\lambda + \frac{1}{2-x_1} + \frac{x_1}{(2-x_1)^2}, \qquad 0 = \frac{dL}{dx_2} = -\lambda + \frac{1}{1-x_2} + \frac{x_2}{(1-x_2)^2},$$

$$0 = -x_1 - x_2 + 2.$$

[2 marks]

c) Solving the equation $0 = \frac{dL}{dx_1}$ for x_1 yields

$$x_1 = \pm 2 + \sqrt{\frac{2}{\lambda}},$$

and solving the equation $0 = \frac{dL}{dx_2}$ for x_2 yields

$$x_2 = 1 \pm \sqrt{\frac{1}{\lambda}}.$$

Using these equations in the third necessary condition yields

"++" $0 = -1 - \frac{\sqrt{2}+1}{\sqrt{\lambda}}$, which does not have any solution;

"+−" $0 = -1 - \frac{\sqrt{2}-1}{\sqrt{\lambda}}$, which does not have any solution;

"−+" $0 = -1 + \frac{\sqrt{2}-1}{\sqrt{\lambda}}$, yielding $\lambda = 3 - 2\sqrt{2}$;

"−−" $0 = -1 + \frac{\sqrt{2}+1}{\sqrt{\lambda}}$, yielding $\lambda = 3 + 2\sqrt{2}$.

Answers

In summary we have two candidate optimal solutions

$$\left(-\sqrt{2}, 2+\sqrt{2}\right), \qquad \left(\sqrt{2}, 2-\sqrt{2}\right).$$

[8 marks]

d) Of the two candidate optimal solutions determined in part c) only one is feasible, namely $x^\star = \left(\sqrt{2}, 2-\sqrt{2}\right)$, which coincides with the solution determined in part a). [2 marks]

e) Note that

$$\nabla^2 L(x^\star) = (1+\sqrt{2})^3 \begin{bmatrix} \sqrt{2} & 0 \\ 0 & 2 \end{bmatrix},$$

which is positive definite, hence $x^\star$ is a local minimiser. [4 marks]

(Note that since $\nabla^2 L(x^\star)$ is positive definite, one does not have to compute the Jacobian of the active constraint and its kernel to verify the second order sufficient condition.)

Answers

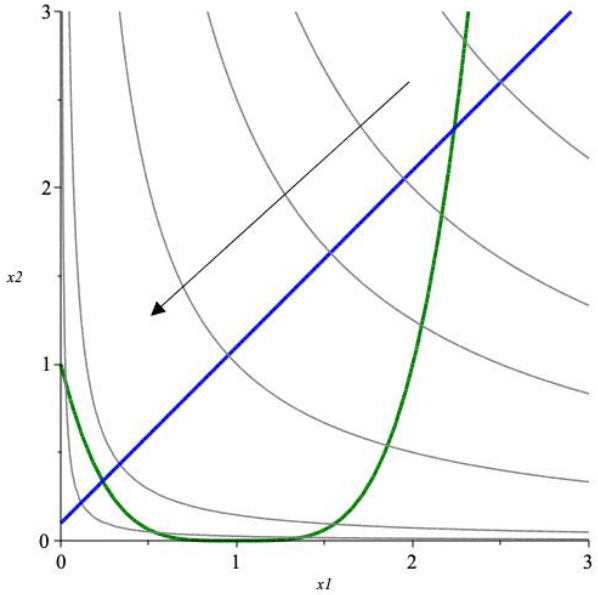
3. a) Solving the equality constraint for x_3 yields $x_3 = x_1 - x_2$, which replaced in the cost and in the second inequality constraint (the first inequality constraint does not contain x_3, hence it is not changed) gives, respectively,

$$x_1^2 + x_2^2 - (x_1 - x_2)^2 = 2x_1x_2,$$

$$-x_2 + (x_1 - 1)^4,$$

thus one obtains the given reduced problem. [2 marks]

b) The admissible set is the set between the blue and green lines in the graph below. The level lines are indicated in grey, with the direction of the arrow in the direction of decreasing values of the objective function.



The optimal point is located at $x_1 = 1$ and $x_2 = 0$, for which the cost is equal to 0. Note finally, that

$$\frac{\partial h_1}{\partial x} = \begin{bmatrix} -1 & 1 \end{bmatrix} \qquad \frac{\partial h_2}{\partial x} = \begin{bmatrix} \star & -1 \end{bmatrix},$$

with $\star = 4(x_1 - 1)^3$. These two vectors are always nonzero and are linearly independent at the points in which both constraints are active, hence all points are regular points for the constraints. [6 marks]

c) The Lagrangian for the problem is

$$L(x, \rho) = 2x_1x_2 + \rho_1 \left(-x_1 + x_2 - \frac{1}{10} \right) + \rho_2 \left(-x_2 + (x_1 - 1)^4 \right).$$

The necessary conditions of optimality are

$$0 = \frac{\partial L}{\partial x_1} = 2x_2 - \rho_1 + 4\rho_2(x_1 - 1)^3, \qquad 0 = \frac{\partial L}{\partial x_2} = 2x_1 + \rho_1 - \rho_2,$$

$$-x_1 + x_2 - \frac{1}{10} \leq 0, \qquad -x_2 + (x_1 - 1)^4 \leq 0, \qquad \rho_1 \geq 0, \qquad \rho_2 \geq 0.$$

Replacing $x_1 = 1$ and $x_2 = 0$ yields $\rho_1 = 0$ and $\rho_2 = 2$, which are therefore the optimal multipliers. At the optimal point, *i.e.* $x^\star = (1, 0)$ and $\rho^\star = (0, 2)$,

$$-x_1^\star + x_2^\star - \frac{1}{10} < 0 \Rightarrow \rho_1^\star = 0, \qquad -x_2^\star + (x_1^\star - 1)^4 = 0 \Rightarrow \rho_2^\star = 2 > 0,$$

hence the condition of strict complementarity holds. [6 marks]

ANSWERS

d) Note that

$$\nabla^2 L(x^\star, \rho^\star) = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$$

and (recall that the second inequality constraint is active at the optimal point)

$$\frac{\partial h_2}{\partial x}(x^\star) = \begin{bmatrix} 0 & -1 \end{bmatrix}.$$

Hence, all nonzero vectors orthogonal to $\frac{\partial h_2}{\partial x}(x^\star)$ are given by $s = \begin{bmatrix} \alpha & 0 \end{bmatrix}^\top$, with $\alpha \neq 0$, yielding

$$s^\top \nabla^2 L(x^\star, \rho^\star) s = 0.$$

The second order sufficient condition of optimality does not hold, hence it cannot be used to establish (local) optimality of $x^\star$. [4 marks]

e) To establish local optimality of $x^\star$ one could consider the restriction of the objective function to the active constraint, that is

$$2x_1 x_2\big|_{x_2=(x_1-1)^4)} = 2x_1(x_1-1)^4,$$

for x_1 in the admissible set. This function has a global minimiser at $x_1 = 1$, which does not satisfy second order sufficient conditions of optimality.

[2 marks]

Answers

4. a) Recall that $\sqrt{2} = 1.414213562$. Note that $f' = 2x$, hence Newton's iteration is

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{x_n^2 + 2}{2x_n} = N(x_n).$$

Running three iterations from $x_0 = 1$ yields

$$x_1 = 1.5, \qquad x_2 = 1.41\dot{6}, \qquad x_3 = 1.414215686.$$

[2 marks]

b) Note that $g(x) = \left((x + (x^2-2))^2 - (x^2-2)\right)/(x^2-2) = x^2 + 2x - 2$, hence Steffensen's iteration is

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{x_n^2 + 2x_n - 2} = \frac{x_n^3 + x_n^2 - 2x_n + 2}{x_n^2 + 2x_n - 2} = S(x_n).$$

Running three iterations from $x_0 = 1$ yields

$$x_1 = 2, \qquad x_2 = 1.\dot{6}, \qquad x_3 = 1.477477478.$$

[4 marks]

c) Note that $Lf(x) = \left(x^2 - 2\right)/(2x^2)$, hence the Chebyshev–Halley iteration is

$$x_{n+1} = \frac{x_n(x_n^2 + 6)}{3x_n^2 + 2} = CH(x_n).$$

Running three iterations from $x_0 = 1$ yields

$$x_1 = 1.4, \qquad x_2 = 1.414213198, \qquad x_3 = 1.414213563.$$

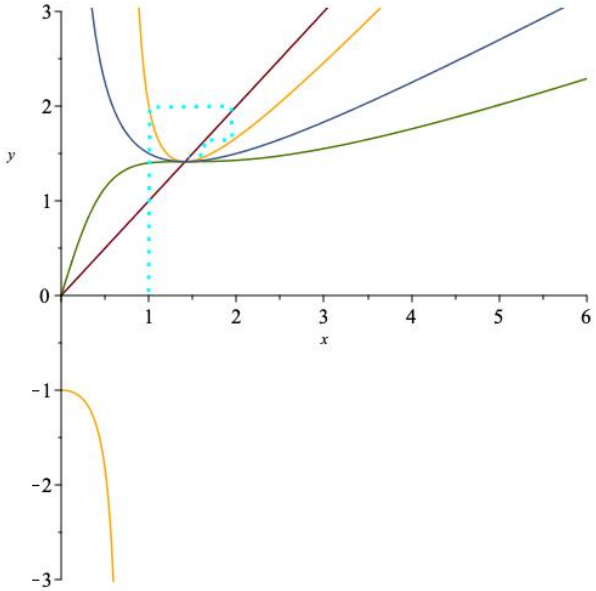
[4 marks]

d) The Chebyshev–Halley iteration yields a very accurate approximation of $\sqrt{2}$ in two steps, much faster than Newton's method, which is reasonable since it uses a more complex iteration and the second derivative of f. Steffensen's iteration has a slower convergence rate, when compared with Newton's, which is again reasonable since it does not use any information on the derivative of f.

(One could also observe that $N'(\sqrt{2}) = S'(\sqrt{2}) = CH'(\sqrt{2}) = 0$, which explains local (quadratic convergence) and that $S''(\sqrt{2}) > N''(\sqrt{2}) > CH''(\sqrt{2}) = 0$, which explains the cubic convergence speed of the Chebyshev–Halley iteration, and the fact that Newton's is faster than Steffensen's.) [2 marks]

Answers

e) The graphs of the functions N (blue), S (orange), CH (green) and x (red) are given in the figure below. The light blue dashed line indicates how Steffensen's iteration works (similar constructions apply to the other iterations).



i) $\sqrt{2}$ is a fixed point of each iteration by construction, since it is the solution of the equation x^2-2. Hence $\sqrt{2}=N(\sqrt{2})=S(\sqrt{2})=CH(\sqrt{2})$. [2 marks]

ii) Similarly to the process described by the light blue dashed line, for any $x_0>0$ Newton's iteration yields a sequence which converges to $\sqrt{2}$. The same applies to Steffensen's iteration provided x_0 is such that $x_1>0$, which is the case for $x_0>\sqrt{3}-1$, and to the Chebyshev–Halley iteration, for any $x_0>0$. [3 marks]

iii) Newton's iteration is not defined for $x_0=0$. Steffensen's iteration is not defined for $x_0=\sqrt{3}-1$ and the iteration does not converge to $\sqrt{2}$ for $x_0\in[0,\sqrt{3}-1)$. The Chebyshev–Halley iteration is well defined for all $x_0\geq 0$, but it has a second fixed point at $x=0$, thus if $x_0=0$, then $x_n=0$, for all $n\geq 0$. [3 marks]

Answers